COMPILERS
(LECTURE 5)

By :

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Minimization of DFA states

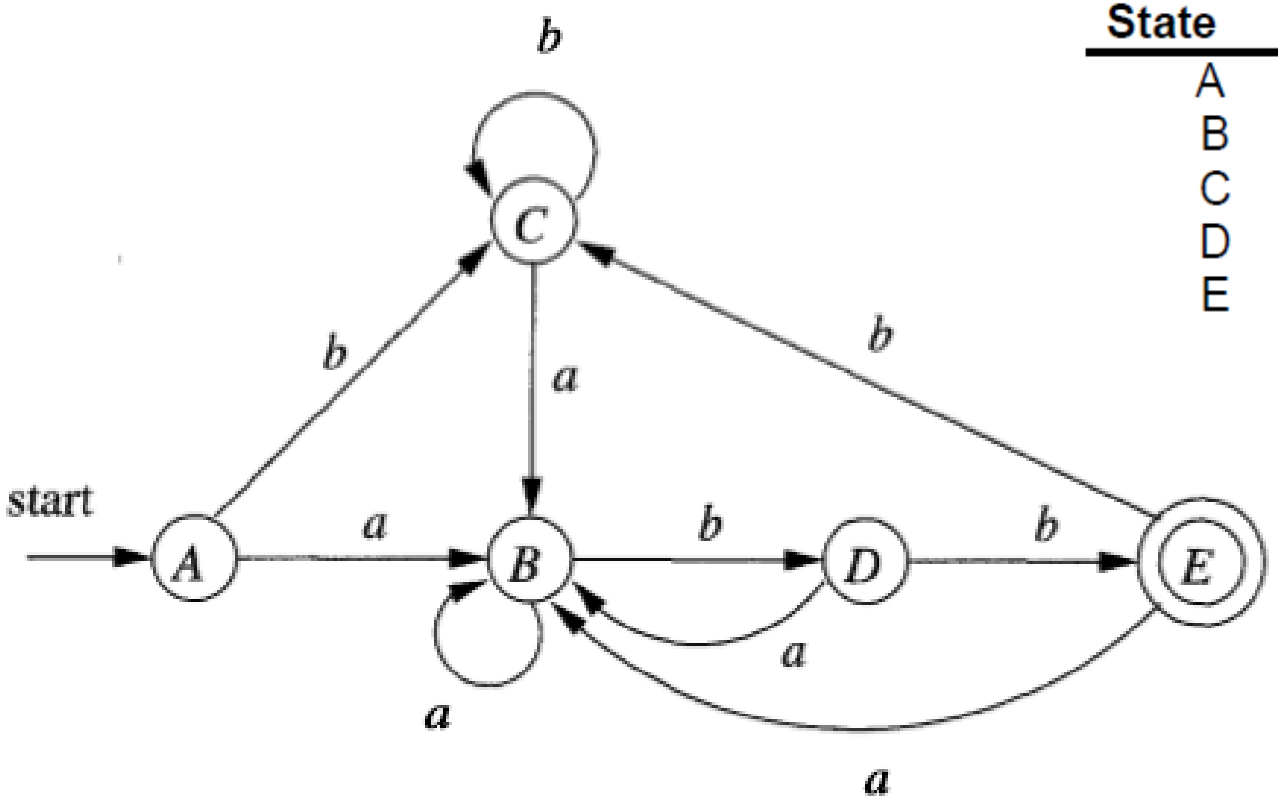
- We split the states into two groups
  - accepting and non-accepting states
- For each group G, split into two subgroups :

  G1 : contain states transitions on a symbol to states in G itself

  G2: contain transitions on a symbol to states outside G

  repeat the above step until no more split can be done

Result of applying the set construction to fig



State	Input **a**	Input **b**
A	B	C
B	B	D
C	B	C
D	B	E
E	B	C

- Initial partitioning : {A,B ,C ,D} set of non-accepting states and {E} set of accepting or final states,

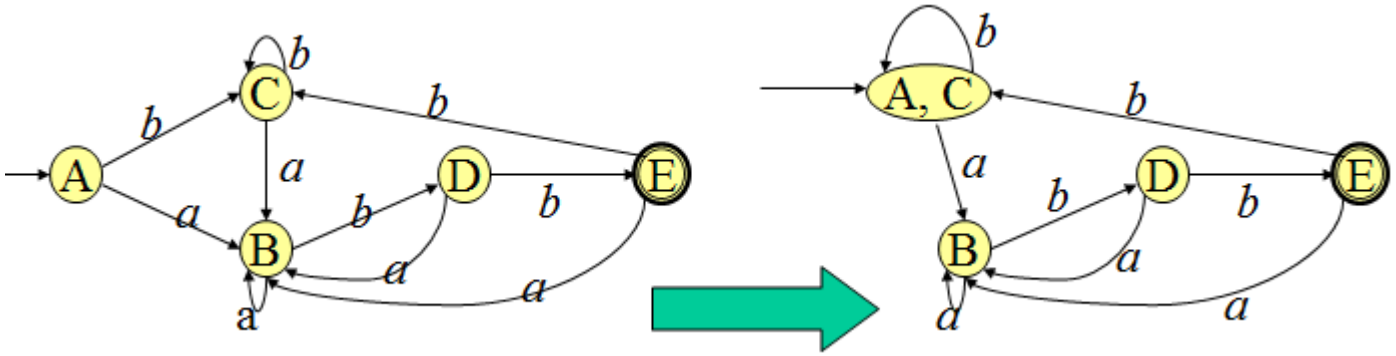
Partitions	Split on a	split on b
{A,B,C ,D} {E}	{A,B ,C,D} {E}	{A,B,C} {D} {E}
{A,B,C} {D} {E}	no split , A,B,C on a goto states in the group	{A, C} {B} {D} {E}
{A, C} {B} {D} {E}	No Split	No Split

A,B,C on b goto states in the group but D go to E

A, C go to members in the group but B goto D

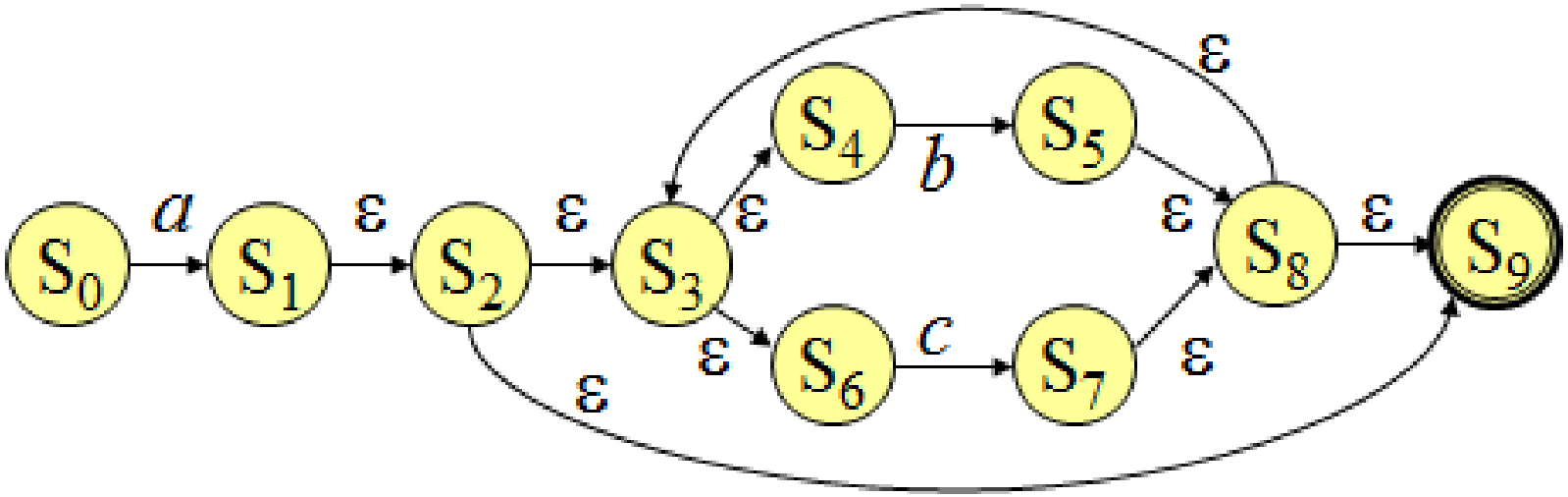
- New Transition Table

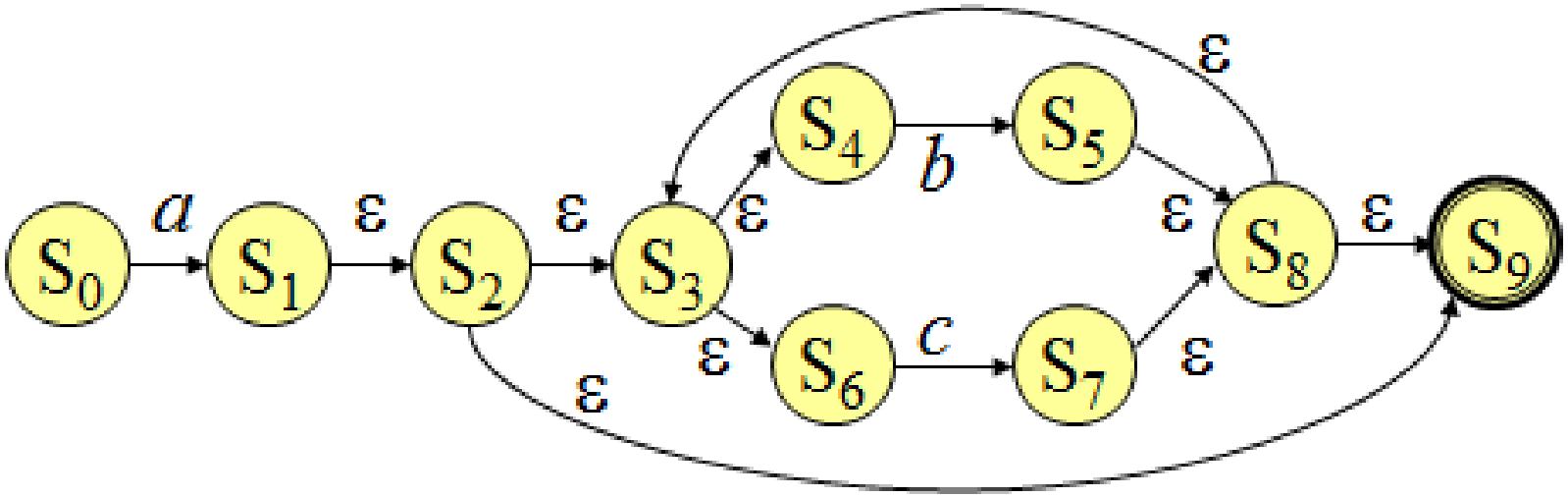
STATE	a	b
AC	B	AC
B	B	D
D	B	E
E	B	AC



Another example

- Language : *a (b* | *c)**
- NFA :

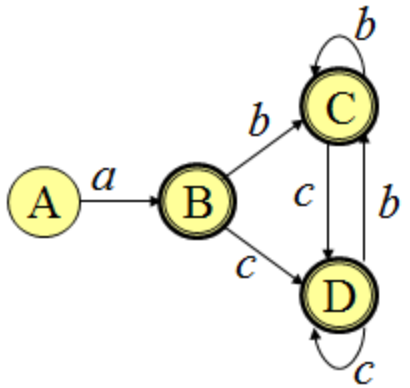




DFA states	NFA states	ε-closure(move(s, x))		
		a	*b*	*c*
A	S0	S1,S2,S3, S4,S6,S9	None	None
B	S1,S2,S3, S4,S6,S9	None	S5,S8,S9, S3,S4,S6	S7,S8,S9, S3,S4,S6
C	S5,S8,S9, S3,S4,S6	None	S5,S8,S9, S3,S4,S6	S7,S8,S9, S3,S4,S6
D	S7,S8,S9, S3,S4,S6	None	S5,S8,S9, S3,S4,S6	S7,S8,S9, S3,S4,S6

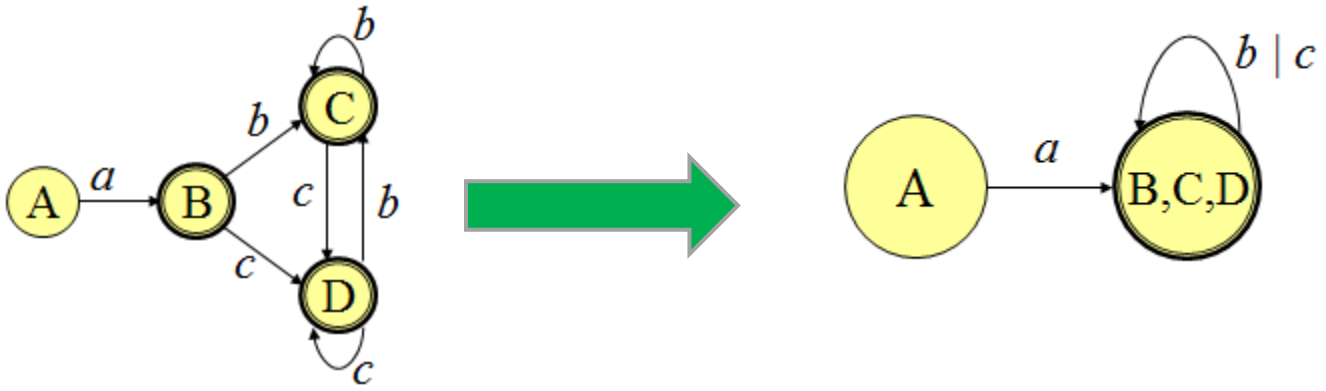
Resulting DFA

DFA states	*a*	*b*	*c*
A	B	None	None
B	None	C	D
C	None	C	D
D	None	C	D



Minimized DFA
{A} : non-final , {B,C,D} :final

	Current groups	Split on *a*	Split on *b*	Split on *c*
0	{B, C, D} {A}	None	None	None



Leftmost and right most derivations

- In leftmost derivations, the leftmost nonterminal in each sentential is always chosen to be replaced .

- In rightmost derivations, the rightmost nonterminal in each sentential is always chosen to be replaced .

Example

- Find leftmost, rightmost derivation & parse tree for: x-2*y if P as follows:
  - **Stat →Expr**
  - **Expr → Expr op Expr| Number| Id**
  - **Op →+| -| * | /**
  - **Id →x | y**
  - **Number →0|1|2|3|4|5|6…….|9**
- **Leftmost**
  - **Stat→ Expr→ Expr Op Expr→ Id Op Expr→x Op Expr→ x - Expr →x- Expr Op Expr → x - Number Op Expr→ x- 2 OP Expr→x-2 * Expr→ x-2* Expr→x-2* ID → x-2*y**

Example cont.

- **Rightmost derivation**
- **Stat→ Expr →Expr Op Expr→ Expr Op Expr Op Expr →Expr Op Expr Op Id→ Expr OP Expr Op y→ Expr Op Expr *y→ Expr OP Number * y → Expr OP 2*y → Expr -2*y → Id-2*y → x-2* y**

Another example

- S→aAS| a
- A→SbA|SS | ba
- Consider string aabbaa
- Leftmost : S➔aAS➔aSbAS➔aabAS➔aabbaS ➔ aabbaa
- Rightmost : S➔aAS ➔ aA a➔a SbA a➔aSbbaa➔aabbaa

Thank you

